

# DRAWING LIST FOR TEST PACKAGE

## D-580

PROJECT NO.: 22328D

PROJECT TITLE : M15X Ph2 Project

| No | PKG No | UNIT | LINE NUMBER          |        |     | P&ID                    |     | REMARK |
|----|--------|------|----------------------|--------|-----|-------------------------|-----|--------|
|    |        |      | ISO DRAWING No       | SHEET  | REV | P&ID No                 | REV |        |
| 1  | 580    | 3FA  | 80A-D-ST1-N-9961-01  | 1 OF 1 | 0   | CJ-M15W14-PR-A20-000113 | 2A  |        |
| 2  | 580    | 3FA  | 80A-D-ST1-N-9962-01  | 1 OF 1 | 0   | CJ-M15W14-PR-A20-000113 | 2A  |        |
| 3  | 580    | 3FA  | 150A-D-ST1-N-9511-01 | 1 OF 1 | 1   | CJ-M15W13-PR-A20-000084 | 2   |        |
| 4  | 580    | 3FA  | 150A-D-ST1-N-9513-01 | 1 OF 1 | 0   | CJ-M15W13-PR-A20-000084 | 2   |        |
| 5  | 580    | 3FA  | 150A-D-ST1-N-9514-01 | 1 OF 1 | 0   | CJ-M15W13-PR-A20-000084 | 2   |        |
| 6  | 580    | 3FA  | 150A-D-ST1-N-9515-01 | 1 OF 1 | 1   | CJ-M15W13-PR-A20-000084 | 2   |        |
| 7  | 580    | 3FA  | 250A-D-ST1-N-9651-01 | 1 OF 1 | 1   | CJ-M15W13-PR-A20-000084 | 2   |        |
| 8  | 580    | 3FA  | OD63-D-HP1-N-9510    | 1 OF 1 | 0   | CJ-M15W13-PR-A20-000084 | 2   |        |
| 9  | 580    | 3FA  | OD63-D-HP1-N-9512    | 1 OF 1 | 0   | CJ-M15W13-PR-A20-000084 | 2   |        |
|    |        |      |                      |        |     |                         |     |        |
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|    |        |      |                      |        |     |                         |     |        |
|    |        |      |                      |        |     |                         |     |        |



|          |                                                |
|----------|------------------------------------------------|
| ITEM NO. | W14-AD-4229 B                                  |
| SERVICE  | CIF MBR 2 <sup>nd</sup> OXIC TANK AIR DIFFUSER |
| Q'TY     | 110                                            |
| TYPE     | MEMBRANE (DISC)                                |
| CAPA.    | L/min.ea                                       |
| MAT'L    | PP + EPDM                                      |

|          |                                   |
|----------|-----------------------------------|
| ITEM NO. | W14-T-4229 B                      |
| SERVICE  | CIF MBR 2 <sup>nd</sup> OXIC TANK |
| Q'TY     | 1                                 |
| TYPE     | RECTANGULAR                       |
| SIZE     | W x L x H m                       |
| CAPA.    | m3                                |
| HRT      | min                               |
| MAT'L    | R.C + FRP                         |

|          |              |
|----------|--------------|
| ITEM NO. | W14-T-4230 B |
| SERVICE  | CIF MBR TANK |
| Q'TY     | 1            |
| TYPE     | RECTANGULAR  |
| SIZE     | W x L x H m  |
| CAPA.    | m3           |
| HRT      | hr           |
| MAT'L    | P.C + FRP    |

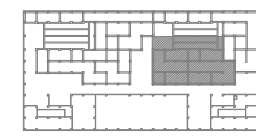
|          |                       |
|----------|-----------------------|
| ITEM NO. | W14-T-4232 B          |
| SERVICE  | CIF MBR CLEANING TANK |
| Q'TY     | 1                     |
| TYPE     | RECTANGULAR           |
| SIZE     | W x L x H m           |
| CAPA.    | m3                    |
| HRT      | min                   |
| MAT'L    | C.S + FRP             |

|          |                      |
|----------|----------------------|
| ITEM NO. | W14-AD-4230 B        |
| SERVICE  | CIF MBR AIR DIFFUSER |
| Q'TY     | ea                   |
| TYPE     | MEMBRANE (DISC)      |
| CAPA.    | L/min.ea             |
| MAT'L    | PP + EPDM            |

|          |                      |
|----------|----------------------|
| ITEM NO. | W14-EJ-4231 B        |
| SERVICE  | CIF MBR EJECTOR UNIT |
| Q'TY     | ea                   |
| TYPE     | BY VENDOR            |
| CAPA.    | N/s.ea               |
| MAT'L    | BY VENDOR            |

|          |                  |
|----------|------------------|
| ITEM NO. | W14-MBR-4230 B   |
| SERVICE  | CIF MBR MEMBRANE |
| CAPA.    | m3/d             |
|          | 15,000           |

대외비



KEY PLAN (3F)

NOTE

1. TANK의 HWL 이하로 연결되는 모든 배관은 BLOCK 밸브 설치할 것.
2. 2m 이상 높은 곳에 설치된 밸브는 CHAIN VALVE 또는 PLATFORM 구성할 것.
3. MANHOLE SIZE : 1000W x 1000L x 2EA/TANK MANHOLE COVER : FRP
4. 운전 시 CLOSE 되는 배관에는 LOTO 설치.
5. CENTRIFUGAL PUMP  
MAT'L 표기 : CASING / IMPELLER
6. 4부 FRP LINING TANK SLEEVE는 SUS304 + FRP LINING 적용.
7. PACKAGE VENDOR 선정 후 4부 SYSTEM (공회 EQUIPMENT SPEC., LINE SIZE, MBR Nos., INSTRUMENT CONTROL SCHEME 등) 확정 예정.
8. 안전포기.
9. 축전포기.
10. 용량계 설치.  
DESIGN RANGE : -1~1 BAR

INTERLOCK NO. 1-4231 B (W14-P-4231 B1/B2 AUTO STOP)  
RENOTE STATUS에 W14-T-4229 B LEVEL (LI-W2321) LOW LOW STOP.

12. OPERATION TIME CHART

| PROCESS   | SERVICE | B/W READY | B/W  | RESTART |
|-----------|---------|-----------|------|---------|
| TIME(sec) | 24hr    | 2         | 30   | 30      |
| XV-2331A  | OPEN    | -         | -    | -       |
| XV-2331B  | -       | -         | OPEN | -       |
| XV-2321   | -       | OPEN      | -    | OPEN    |
| B/W PUMP  | -       | -         | RUN  | -       |

11. 나머지 ON-OFF VALVE들도 동일하게 적용한다.
13. CIF MBR TANK CRANE를 PACKAGE VENDOR에게 공급한다.
14. DELETED.
15. INSTRUMENT VENDOR 선정 후 CONTROL SCHEME 및 CONTROL VALVE INLET / OUTLET NOZZLE SIZE 확정 예정.
16. MANHOLE 2개소 설치하여야 한다.
17. SUPPLIED BY MBR PACKAGE VENDOR.
18. HIGH POINT에 SIPHON BREAKER 설치할 것.  
(CHECK VALVE의 경우 DUAL PLATE TYPE으로 적용)
19. 개개의 세정용 CASSETTE CONNECTION 용도.  
MBR CLEANING TANK 인근에 설치할 것.
20. 유체의 흐름을 확인할 수 있도록 LINE 상에 PRESSURE GAUGE를 설치할 것. (TAG NO. 아래 참조)

- 150A-AIR-ST1-N-9528 : PG-W2381
21. CIF MBR CLEANING TANK SHALL BE INSTALLED COVER WITH DUCT. OVERFLOW LINE(80A) TO DRAIN HEADER.
22. 하도개 설치를 위한 FUTURE CONNECTIONS.
23. PORT SIZE 350A. (TEE + 단관 + CAP 마킹 : 단관 길이는 W14-P-4231 PUMP의 CASING 일단 보다 높게 제작)
24. AIR RELEASE VALVE
25. SET POINT : 6BAR ~ 3 ~ 1BAR
26. SIZE IS 3,500mm x 1,600mm

|      |             |          |
|------|-------------|----------|
| 2A   | REVISED AFC | 25.08.14 |
| 2    | REVISED AFC | 25.01.17 |
| 1    | REVISED AFC | 24.10.31 |
| 수정   | 실명          | 날짜       |
| REV. | DESCRIPTION | DATE     |

도면명  
DWG. TITLE  
PIPING & INSTRUMENT DIAGRAM  
CIF MBR DISCHARGE & CIRCULATION PUMP B (2/2)

SK 에코플랜트

사명  
PROJECT TITLE

M15X Ph4 WWT Project

도면명  
DWG. TITLE

PIPING & INSTRUMENT DIAGRAM

CIF MBR DISCHARGE & CIRCULATION PUMP B (2/2)

척  
SCALE

NONE

날짜  
DATE

2025. 08. 14

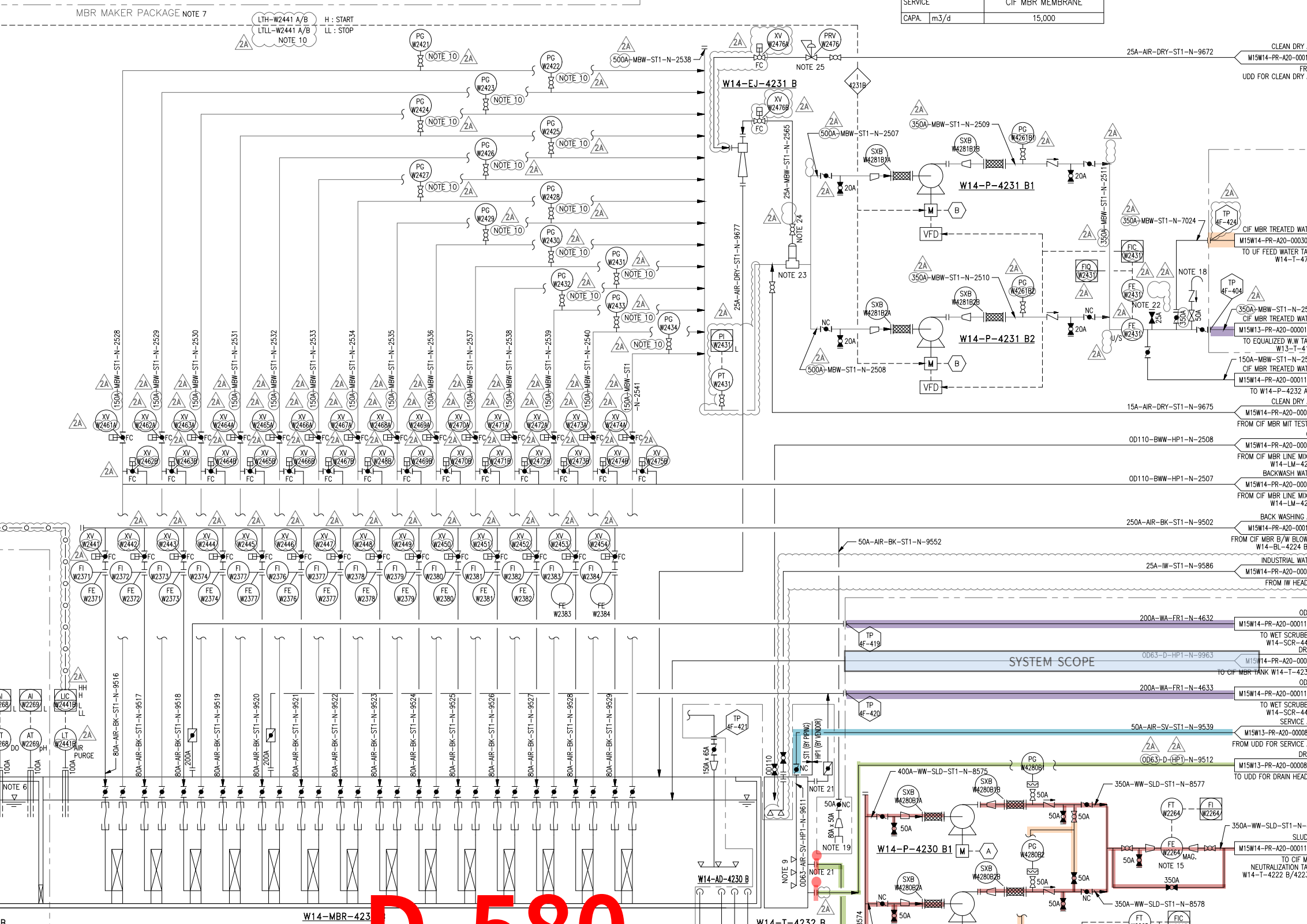
PROJECT NO. 22330D

REV. 2A

|          |                      |
|----------|----------------------|
| ITEM NO. | W14-P-4231 B1/B2     |
| SERVICE  | CIF MBR SUCTION PUMP |
| Q'TY     | 1 + 1                |
| TYPE     | CENTRIFUGAL (VWF)    |
| CAPA.    | m3/hr x mH           |
| MAT'L    | SSC13/SSC13          |
| POWER kW | (132)                |

|          |                            |
|----------|----------------------------|
| ITEM NO. | W14-P-4230 B1/B2           |
| SERVICE  | CIF MBR RECIRCULATION PUMP |
| Q'TY     | 1 + 1                      |
| TYPE     | CENTRIFUGAL                |
| CAPA.    | m3/hr x mH                 |
| MAT'L    | SSC13/SSC13                |
| POWER kW | (90)                       |

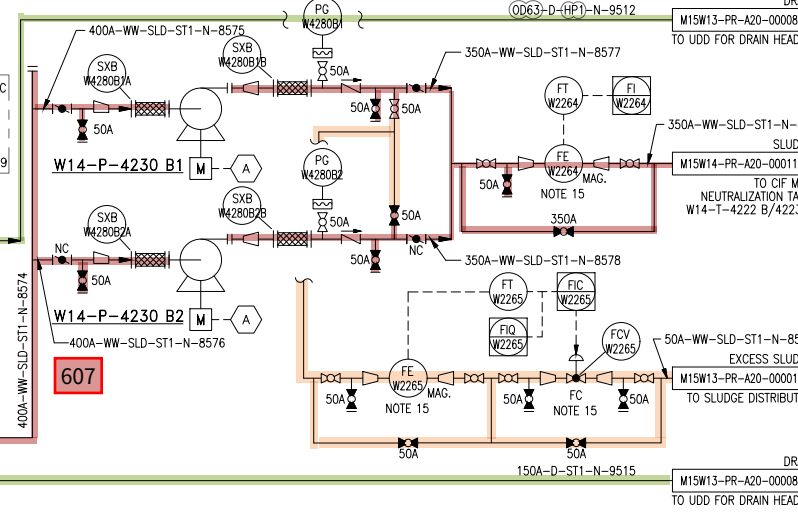
D-580



W14-T-4229 B  
NOTE 16

W14-MBR-4230 B  
NOTE 17

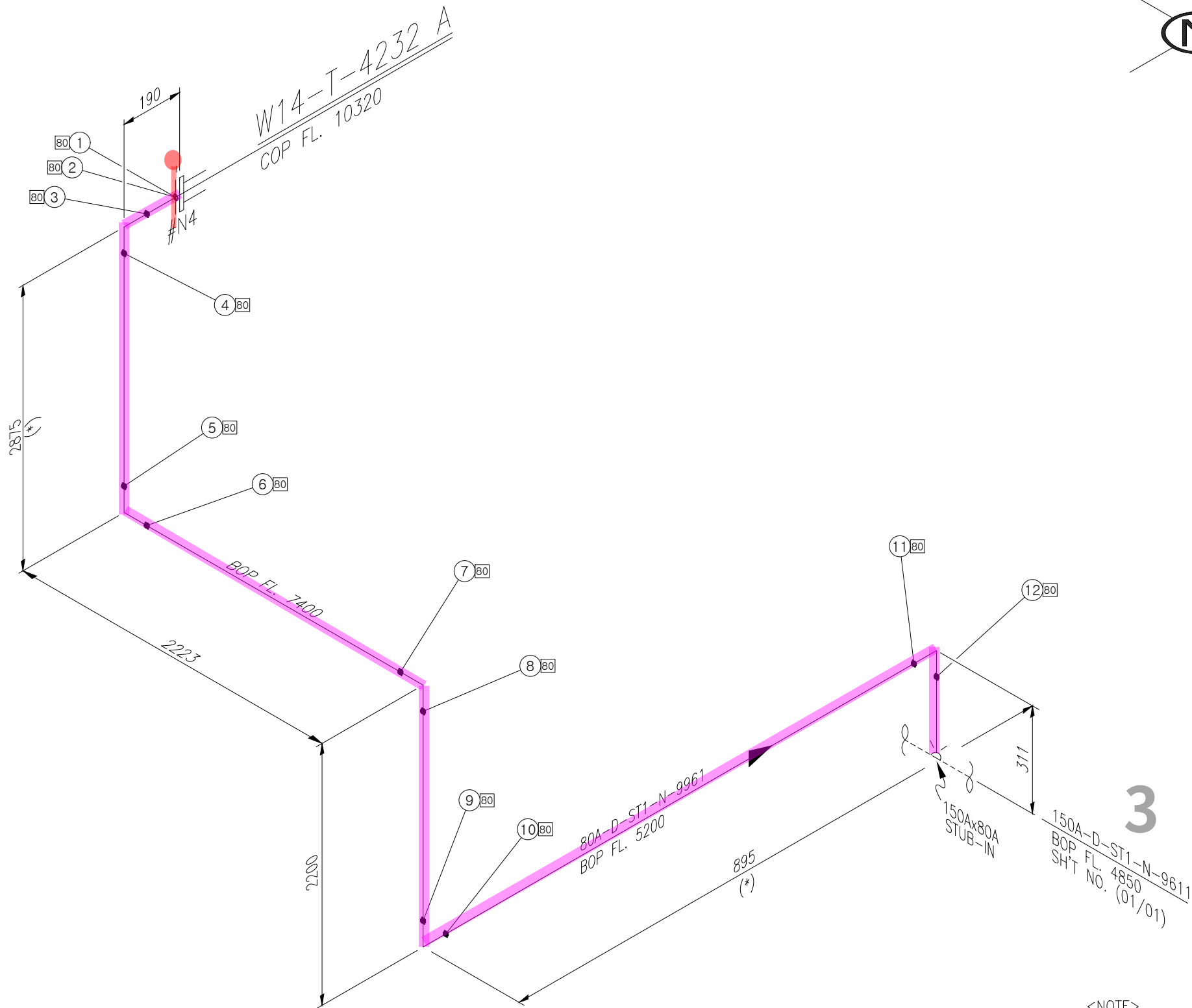
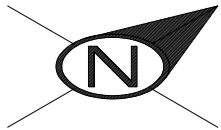
W14-T-4232 B  
NOTE 17



C:\Users\WTOP\Desktop\W&ID\251013\W[M15X Ph-4 for 인허가]\CAD FILE\0910\WCJ-M15W14-PR-A20-000115 R1 P&ID FOR CIF M



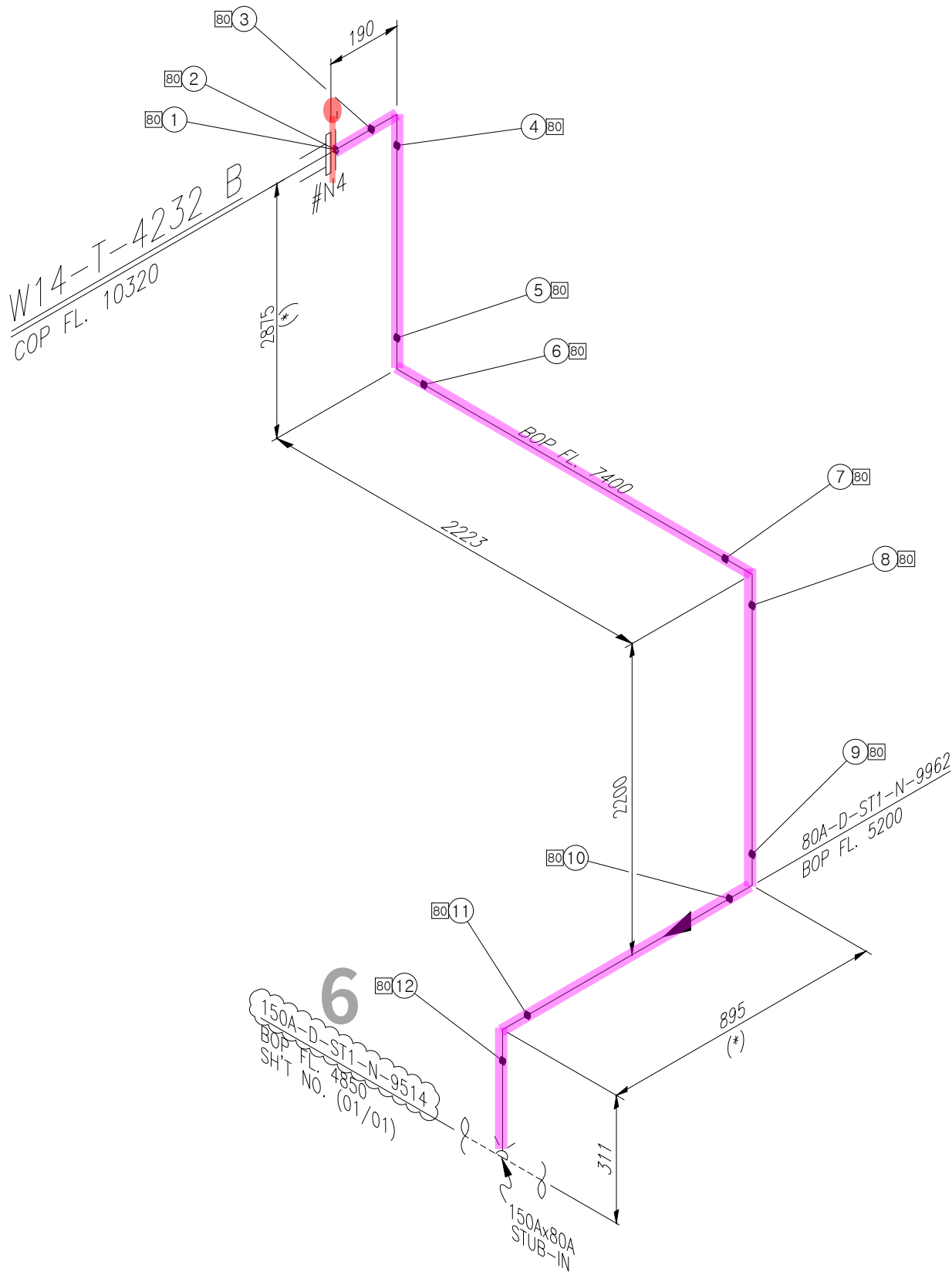
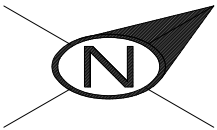




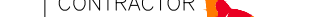
\* PIPE : P/P1 \* 90° ELBOW : E90 \* 45° ELBOW : E45 \* TEE : T \* RED.TEE : TR \* STUB-IN : STU \* REINFORCING PAD : RP \* CON.RED : RC \* ECC.RED : RE \* CON.SWAGE : RCS/RCS1 \* ECC.SWAGE : RES/RES1 \* NIPPLE : NP/NP1/NP2 \* CAP : CAP/CAP1 \* PLUG : PLG \* FULL COUPLING : FC \* HALF COUPLING : HF/HF1  
\* STUB-IN : STU \* SOCKET(HDPE/PVC) : SC \* SADDLE(HDPE) : SD \* FLANGE : F/F1 \* GASKET : G/G1/G2 \* BOLT & NUT : B/B1 \* BLIND FLANGE : FB/FB1 \* GATE VA. : VG/VG1/VG2 \* BALL VA. : VB/VBLO/VB1/VB2LO \* BUTTERFLY VA. : VF/VF4 \* CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7  
\* GLOBE VA. : VL/VL1/VL2 \* DIAPHRAGM VA. : VD/VDLO \* DAMPER : VP \* QUICK COUPLING : SQC \* FLEXIBLE JOINT(STEEL) : FLXB \* FLEXIBLE JOINT(RUBBER) : FLXR \* FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL \* FLEXIBLE HOSE : FLXH \* STEAM TRAP : ST \* SPRAY NOZZLE : SPN \* EMERGENCY SHOWER : SSH  
\* STRAINER(Y-TYPE) : YSTR \* STRAINER(T-TYPE) : TSTR \* STRAINER(BUCKET-TYPE) : BSTR \* ANCHOR PLATE : STOP \* BIRD SCREEN : BS \* ISOLATION KIT : IK \* UNLOADING BOX : UB \* AIR TRAP : AT \* LOTO : LOTO \* SIGHT GLASS : SG \* PRESSURE GAUGE : PG \* SHILED : SH \* SPECIAL BOLT & NUT : BL/BL1

|            |        |   |           |                     |               |                           |     |                                                                                       |          |                           |     |     |       |                                                                                       |                  |      |
|------------|--------|---|-----------|---------------------|---------------|---------------------------|-----|---------------------------------------------------------------------------------------|----------|---------------------------|-----|-----|-------|---------------------------------------------------------------------------------------|------------------|------|
| INSULATION | CODE   | N | CONDITION | PRESS.<br>(kg/cm2g) | TEMP.<br>(°C) | AREA                      | W14 |                                                                                       |          |                           |     |     |       | CONTRACTOR                                                                            | PROJECT NAME     |      |
|            | THICK. | — | DESIGN    | 1.0                 | 60            | UNIT                      | 3FA |                                                                                       |          |                           |     |     |       |  | M15X Ph4 Project |      |
|            |        |   | OPERATING | ATM                 | AMB           | REF. PLAN DWG.            |     |                                                                                       |          |                           |     |     |       | COMMODITY                                                                             | JOB NO.          |      |
|            |        |   |           |                     |               | CJ-M15W14-PI-A30-030002-E |     |  | 25.03.31 | APPROVED FOR CONSTRUCTION | 황래중 | 안남후 | 박동영   | D                                                                                     | 22330D           |      |
|            |        |   |           |                     |               | REF. P & ID               |     | REV. NO.                                                                              | DATE     | DESCRIPTION               | BY  | CHK | APP'D | LINE NO.                                                                              | SH'T NO.         | REV. |
|            |        |   |           |                     |               | CJ-M15W14-PR-A20-000113   |     |                                                                                       |          |                           |     |     |       | 80A-D-ST1-N-9961                                                                      | 01 / 01          | 0    |

경고 : 본 도면은 SK HYNIX의 기밀비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음.



\* PIPE : P/P1 \* 90° ELBOW : E90 \* 45° ELBOW : E45 \* TEE : T \* RED.TEE : TR \* STUB-IN : STU \* REINFORCING PAD : RP \* CON.RED : RC \* ECC.RED : RE \* CON.SWAGE : RCS/RCS1 \* ECC.SWAGE : RES/RES1 \* NIPPLE : NP/NP1/NP2 \* CAP : CAP/CAP1 \* PLUG : PLG \* FULL COUPLING : FC \* HALF COUPLING : HF/HF1  
\* STUB-IN : STU \* SOCKET(HDPE/PVC) : SC \* SADDLE(HDPE) : SD \* FLANGE : F/F1 \* GASKET : G/G1/G2 \* BOLT & NUT : B/B1 \* BLIND FLANGE : FB/FB1 \* GATE VA. : VG/VG1/VG2 \* BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO \* BUTTERFLY VA. : VF/VF4 \* CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7  
\* GLOBE VA. : VL/VL1/VL2 \* DIAPHRAGM VA. : VD/VDLO \* DAMPER : VP \* QUICK COUPLING : SQC \* FLEXIBLE JOINT(STEEL) : FLXB \* FLEXIBLE JOINT(RUBBER) : FLXR \* FLEXIBLE JOINT(STEEL PIPE LINED) : FLXL \* FLEXIBLE HOSE : FLXH \* STEAM TRAP : ST \* SPRAY NOZZLE : SPN \* EMERGENCY SHOWER : SSH  
\* STRAINER(Y-TYPE) : YSTR \* STRAINER(T-TYPE) : TSTR \* STRAINER(BUCKET-TYPE) : BSTR \* ANCHOR PLATE : STOP \* BIRD SCREEN : BS \* ISOLATION KIT : IK \* UNLOADING BOX : UB \* AIR TRAP : AT \* LOTO : LOTO \* SIGHT GLASS : SG \* PRESSURE GAUGE : PG \* SHILED : SH \* SPECIAL BOLT & NUT : BL/BL1

|            |        |   |           |                     |               |                                             |     |                                                                                       |          |                           |     |     |       |                                                                                       |                  |                     |           |
|------------|--------|---|-----------|---------------------|---------------|---------------------------------------------|-----|---------------------------------------------------------------------------------------|----------|---------------------------|-----|-----|-------|---------------------------------------------------------------------------------------|------------------|---------------------|-----------|
| INSULATION | CODE   | N | CONDITION | PRESS.<br>(kg/cm2g) | TEMP.<br>(°C) | AREA                                        | W14 |                                                                                       |          |                           |     |     |       | CONTRACTOR                                                                            | PROJECT NAME     |                     |           |
|            | THICK. | - | DESIGN    | 1.0                 | 60            | UNIT                                        | 3FA |                                                                                       |          |                           |     |     |       |  | M15X Ph4 Project |                     |           |
|            |        |   | OPERATING | ATM                 | AMB           | REF. PLAN DWG.<br>CJ-M15W14-PI-A30-030002-E |     |                                                                                       |          |                           |     |     |       | COMMODITY                                                                             | JOB NO.          |                     |           |
|            |        |   |           |                     |               | REF. P & ID<br>CJ-M15W14-PR-A20-000115      |     |  | 25.03.31 | APPROVED FOR CONSTRUCTION | 황래중 | 안남후 | 박동영   | LINE NO.                                                                              | 80A-D-ST1-N-9962 | SH'T NO.<br>01 / 01 | REV.<br>0 |
|            |        |   |           |                     |               |                                             |     | REV. NO.                                                                              | DATE     | DESCRIPTION               | BY  | CHK | APP'D |                                                                                       |                  |                     |           |



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[illegible]

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|                           |                           |                                |                        |                                |                                 |                                           |                         |                                          |                          |                                              |                       |                               |              |                      |                          |
|---------------------------|---------------------------|--------------------------------|------------------------|--------------------------------|---------------------------------|-------------------------------------------|-------------------------|------------------------------------------|--------------------------|----------------------------------------------|-----------------------|-------------------------------|--------------|----------------------|--------------------------|
| * PIPE : P/P1             | * 90° ELBOW : E90         | * 45° ELBOW : E45              | * TEE : T              | * RED.TEE : TR                 | * SUB-IN : STU                  | * REINFORCING PAD : RP                    | * CON.RED : RC          | * ECC.RED : RE                           | * CON.SWAGE : RCS/RCS1   | * ECC.SWAGE : RES/RES1                       | * NIPPLE : NP/NP1/NP2 | * CAP : CAP/CAP1              | * PLUG : PLG | * FULL COUPLING : FC | * HALF COUPLING : HF/HF1 |
| * SUB-IN : STU            | * SOCKET(HDPE/PVC) : SC   | * SADDLE(HDPE) : SD            | * FLANGE : F/F1        | * GASKET : G/G1/G2             | * BOLT & NUT : B/B1             | * BLIND FLANGE : FB/FB1                   | * GATE VA. : VG/VG1/VG2 | * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO | * BUTTERFLY VA. : VF/VF4 | * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7 |                       |                               |              |                      |                          |
| * GLOBE VA. : VL/VL1/VL2  | * DIAPHRAGM VA. : VD/VDLO | * DAMPER : VP                  | * QUICK COUPLING : SQC | * FLEXIBLE JOINT(STEEL) : FLXB | * FLEXIBLE JOINT(RUBBER) : FLXR | * FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL | * FLEXIBLE HOSE : FLXH  | * STEAM TRAP : ST                        | * SPRAY NOZZLE : SPN     | * EMERGENCY SHOWER : SSH                     |                       |                               |              |                      |                          |
| * STRAINER(Y-TYPE) : YSTR | * STRAINER(T-TYPE) : TSTR | * STRAINER(BUCKET-TYPE) : BSTR | * ANCHOR PLATE : STOP  | * BIRD SCREEN : BS             | * ISOLATION KIT : IK            | * UNLOADING BOX : UB                      | * AIR TRAP : AT         | * LOTO : LOTO                            | * SIGHT GLASS : SG       | * PRESSURE GAUGE : PG                        | * SHIELDED : SH       | * SPECIAL BOLT & NUT : BL/BL1 |              |                      |                          |

[illegible]

[illegible]

<NOTE>  
1. (\*) FIELD ADJUST

|                           |                           |                                |                        |                                |                                 |                                           |                         |                                    |                          |                                              |                       |                               |              |                      |                          |
|---------------------------|---------------------------|--------------------------------|------------------------|--------------------------------|---------------------------------|-------------------------------------------|-------------------------|------------------------------------|--------------------------|----------------------------------------------|-----------------------|-------------------------------|--------------|----------------------|--------------------------|
| * PIPE : P/P1             | * 90° ELBOW : E90         | * 45° ELBOW : E45              | * TEE : T              | * RED.TEE : TR                 | * STUB-IN : STU                 | * REINFORCING PAD : RP                    | * CON.RED : RC          | * ECC.RED : RE                     | * CON.SWAGE : RCS/RCS1   | * ECC.SWAGE : RES/RES1                       | * NIPPLE : NP/NP1/NP2 | * CAP : CAP/CAP1              | * PLUG : PLG | * FULL COUPLING : FC | * HALF COUPLING : HF/HF1 |
| * STUB-IN : STU           | * SOCKET(HDPE/PVC) : SC   | * SADDLE(HDPE) : SD            | * FLANGE : F/F1        | * GASKET : G/G1/G2             | * BOLT & NUT : B/B1             | * BLIND FLANGE : FB/FB1                   | * GATE VA. : VG/VG1/VG2 | * BALL VA. : VB/VB1/VB10/VB2/VB2L0 | * BUTTERFLY VA. : VF/VF4 | * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7 |                       |                               |              |                      |                          |
| * GLOBE VA. : VL/VL1/VL2  | * DIAPHRAGM VA. : VD/VDL0 | * DAMPER : VP                  | * QUICK COUPLING : SQC | * FLEXIBLE JOINT(STEEL) : FLXB | * FLEXIBLE JOINT(RUBBER) : FLXR | * FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL | * FLEXIBLE HOSE : FLXH  | * STEAM TRAP : ST                  | * SPRAY NOZZLE : SPN     | * EMERGENCY SHOWER : SSH                     |                       |                               |              |                      |                          |
| * STRAINER(Y-TYPE) : YSTR | * STRAINER(T-TYPE) : TSTR | * STRAINER(BUCKET-TYPE) : BSTR | * ANCHOR PLATE : STOP  | * BIRD SCREEN : BS             | * ISOLATION KIT : IK            | * UNLOADING BOX : UB                      | * AIR TRAP : AT         | * LOTO : LOTO                      | * SIGHT GLASS : SG       | * PRESSURE GAUGE : PG                        | * SHILED : SH         | * SPECIAL BOLT & NUT : BL/BL1 |              |                      |                          |

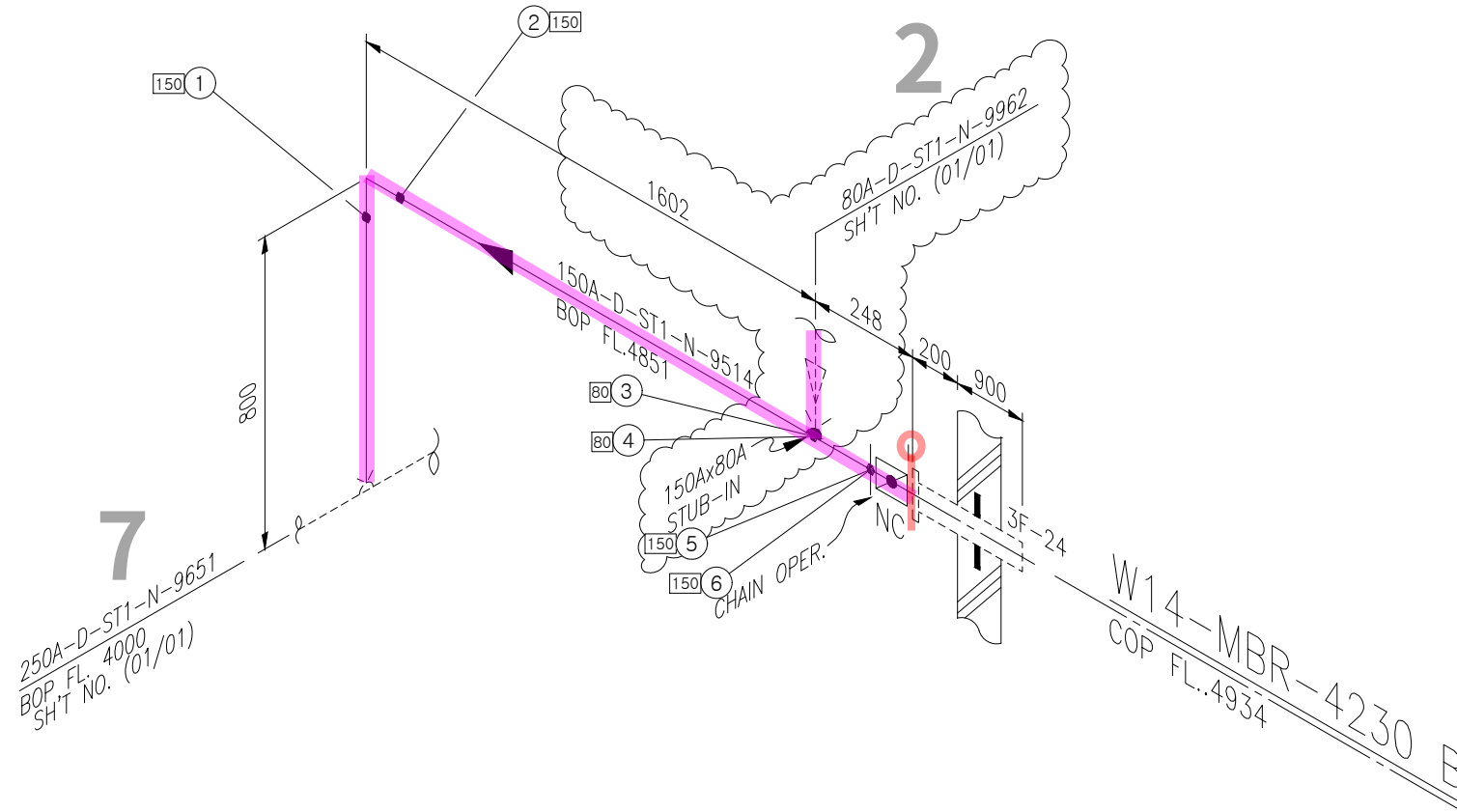
|                                                                                       |  |                                                                                                                                                                             |      |
|---------------------------------------------------------------------------------------|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| CLIENT                                                                                |  |   |      |
| CONTRACTOR                                                                            |  | PROJECT NAME                                                                                                                                                                |      |
|  |  | M15X Ph4 Project                                                                                                                                                            |      |
| COMMODITY                                                                             |  | JOB NO.                                                                                                                                                                     |      |
| D                                                                                     |  | 22330D                                                                                                                                                                      |      |
| LINE NO.                                                                              |  | SH'T NO.                                                                                                                                                                    | REV. |
| 150A-D-ST1-N-9513                                                                     |  | 01 / 01                                                                                                                                                                     | 0    |

|            |        |   |           |                     |                                        |                                             |                                                                                       |             |                           |     |       |     |
|------------|--------|---|-----------|---------------------|----------------------------------------|---------------------------------------------|---------------------------------------------------------------------------------------|-------------|---------------------------|-----|-------|-----|
| INSULATION | CODE   | N | CONDITION | PRESS.<br>(kg/cm2g) | TEMP.<br>(°C)                          | AREA W14                                    |                                                                                       |             |                           |     |       |     |
|            | THICK. | — | DESIGN    | 1.0                 | 60                                     | UNIT 3FA                                    |                                                                                       |             |                           |     |       |     |
|            |        |   | OPERATING | ATM                 | AMB                                    | REF. PLAN DWG.<br>CJ-M15W14-PI-A30-030002-C |  | 25.01.24    | APPROVED FOR CONSTRUCTION | 황래중 | 안남후   | 박동영 |
|            |        |   |           |                     | REF. P & ID<br>CJ-M15W14-PR-A20-000113 |                                             |                                                                                       |             |                           |     |       |     |
|            |        |   |           |                     |                                        | REV. NO.                                    | DATE                                                                                  | DESCRIPTION | BY                        | CHK | APP'D |     |





| ITEM | SIZE     | Q'TY | DESCRIPTION                                      |
|------|----------|------|--------------------------------------------------|
| ST1  |          |      |                                                  |
| P    | 150A     | 2.5M | PIPE, STS 304TP, ERW, BE – SCH 10S               |
| E90  | 150A     | 1EA  | ELBOW 90 DEG LR, STS304, ERW, BE – SCH 10S       |
| F    | 150A     | 1EA  | FLG SO, KS 10K, RF, STS304                       |
| G    | 150A     | 1EA  | GASKET, KS 10K, RF, PTFE+EPDM, 3.0MM TH'K        |
| G1   | 150A     | 1EA  | GASKET, KS 10K, FF, PTFE+EPDM, 3.0MM TH'K        |
| BL   | 150A     | 1SET | MACHINE BOLT & NUTS STS 304 / W WASHERS(SPECIAL) |
| VFC  | 150A     | 1EA  | BUTTERFLY VALVE, KS 10K, WAFER RF, BODY-CCD450 / |
|      |          |      | TRIM-ST304 / EPDM-LINED, CHAIN                   |
| LOTO | 150A     | 1EA  | LOTO                                             |
| STU  | 150Ax80A | 1EA  | STUB-IN BRANCH                                   |



<NOTE>  
1. (\*) FIELD ADJUST

|                           |                           |                                |                        |                                |                                 |                                           |                         |                                         |                          |                                           |                       |                               |              |                      |                          |
|---------------------------|---------------------------|--------------------------------|------------------------|--------------------------------|---------------------------------|-------------------------------------------|-------------------------|-----------------------------------------|--------------------------|-------------------------------------------|-----------------------|-------------------------------|--------------|----------------------|--------------------------|
| * PIPE : P/P1             | * 90° ELBOW : E90         | * 45° ELBOW : E45              | * TEE : T              | * RED.TEE : TR                 | * STUB-IN : STU                 | * REINFORCING PAD : RP                    | * CON.RED : RC          | * ECC.RED : RE                          | * CON.SWAGE : RCS/RCS1   | * ECC.SWAGE : RES/RES1                    | * NIPPLE : NP/NP1/NP2 | * CAP : CAP/CAP1              | * PLUG : PLG | * FULL COUPLING : FC | * HALF COUPLING : HF/HF1 |
| * STUB-IN : STU           | * SOCKET(HDPE/PVC) : SC   | * SADDLE(HDPE) : SD            | * FLANGE : F/F1        | * GASKET : G/G1/G2             | * BOLT & NUT : B/B1             | * BLIND FLANGE : FB/FB1                   | * GATE VA. : VG/VG1/VG2 | * BALL VA. : VB/VBLO/VB1/VB1O/VB2/VB2LO | * BUTTERFLY VA. : VF/VF4 | * CHECK VA. : VC/VC1/VC2/VC3/VC4/VCS/VCS7 |                       |                               |              |                      |                          |
| * GLOBE VA. : VL/VL1/VL2  | * DIAPHRAGM VA. : VD/VDLO | * DAMPER : VP                  | * QUICK COUPLING : SQC | * FLEXIBLE JOINT(STEEL) : FLXB | * FLEXIBLE JOINT(RUBBER) : FLXR | * FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL | * FLEXIBLE HOSE : FLXH  | * STEAM TRAP : ST                       | * SPRAY NOZZLE : SPN     | * EMERGENCY SHOWER : SSH                  |                       |                               |              |                      |                          |
| * STRAINER(Y-TYPE) : YSTR | * STRAINER(T-TYPE) : TSTR | * STRAINER(BUCKET-TYPE) : BSTR | * ANCHOR PLATE : STOP  | * BIRD SCREEN : BS             | * ISOLATION KIT : IK            | * UNLOADING BOX : UB                      | * AIR TRAP : AT         | * LOTO : LOTO                           | * SIGHT GLASS : SG       | * PRESSURE GAUGE : PG                     | * SHILED : SH         | * SPECIAL BOLT & NUT : BL/BL1 |              |                      |                          |

|                                                                                       |  |                                                                                                                                                                             |      |
|---------------------------------------------------------------------------------------|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| CLIENT                                                                                |  |   |      |
| CONTRACTOR                                                                            |  | PROJECT NAME                                                                                                                                                                |      |
|  |  | M15X Ph4 Project                                                                                                                                                            |      |
| COMMODITY                                                                             |  | JOB NO.                                                                                                                                                                     |      |
| D                                                                                     |  | 22330D                                                                                                                                                                      |      |
| LINE NO.                                                                              |  | SH'T NO.                                                                                                                                                                    | REV. |
| 150A-D-ST1-N-9514                                                                     |  | 01 / 01                                                                                                                                                                     | 0    |

|            |        |   |           |                                  |               |                           |     |                                                                                       |          |                           |     |     |       |                                                                                       |                  |      |
|------------|--------|---|-----------|----------------------------------|---------------|---------------------------|-----|---------------------------------------------------------------------------------------|----------|---------------------------|-----|-----|-------|---------------------------------------------------------------------------------------|------------------|------|
| INSULATION | CODE   | N | CONDITION | PRESS.<br>(kg/cm <sup>2</sup> g) | TEMP.<br>(°C) | AREA                      | W14 |                                                                                       |          |                           |     |     |       | CONTRACTOR                                                                            | PROJECT NAME     |      |
|            | THICK. | — | DESIGN    | 1.0                              | 60            | UNIT                      | 3FA |                                                                                       |          |                           |     |     |       |  | M15X Ph4 Project |      |
|            |        |   | OPERATING | ATM                              | AMB           | REF. PLAN DWG.            |     |                                                                                       |          |                           |     |     |       | COMMODITY                                                                             | JOB NO.          |      |
|            |        |   |           |                                  |               | CJ-M15W14-PI-A30-030002-C |     |  | 25.01.24 | APPROVED FOR CONSTRUCTION | 황래중 | 안남후 | 박동영   | D                                                                                     | 22330D           |      |
|            |        |   |           |                                  |               | REF. P & ID               |     | REV.                                                                                  | DATE     | DESCRIPTION               | BY  | CHK | APP'D | LINE NO.                                                                              | SH'T NO.         | REV. |
|            |        |   |           |                                  |               | CJ-M15W14-PR-A20-000115   |     | NO.                                                                                   |          |                           |     |     |       | 150A-D-ST1-N-9514                                                                     | 01 / 01          | 0    |

경고 : 본 문건은 SK HYNIX의 기밀비밀 자료로서, 어떠한 경우에도 무단 복제·배포, 제3자에게 공개, 누설 또는 기타 방법으로 유출하여서는 안 됩니다. 위 사항을 위반한 자에 대해서는 법적 조치를 취할 수 있으며, 회사는 은포자의 귀책사유로 인해 피해가 발생할 경우 은포자에게 민/형사상 책임을 물을 수 있습니다.

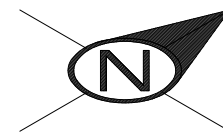


|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |  |        |   |           |                     |               |                |                           |                                                                                       |          |                           |                                                                                                                     |  |     |     |       |            |  |                                                                                       |  |                  |  |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--------|---|-----------|---------------------|---------------|----------------|---------------------------|---------------------------------------------------------------------------------------|----------|---------------------------|---------------------------------------------------------------------------------------------------------------------|--|-----|-----|-------|------------|--|---------------------------------------------------------------------------------------|--|------------------|--|
| <div>* PIPE : P/P1 * 90° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1</div> <div>* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT &amp; NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VBLO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VCS/VCS/V7</div> <div>* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH</div> <div>* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT &amp; NUT : BL/BL1</div> |  |        |   |           |                     |               |                |                           |                                                                                       |          |                           | <div>CLIENT</div> <div>6</div> |  |     |     |       |            |  |                                                                                       |  |                  |  |
| INSULATION                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |  | CODE   | N | CONDITION | PRESS.<br>(kg/cm2g) | TEMP.<br>(°C) | AREA           | W14                       |                                                                                       |          |                           |                                                                                                                     |  |     |     |       | CONTRACTOR |  |  |  | PROJECT NAME     |  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |  | THICK. | — | DESIGN    | 1.0                 | 60            | UNIT           | 3FA                       |                                                                                       |          |                           |                                                                                                                     |  |     |     |       | COMMODITY  |  | D                                                                                     |  | M15X Ph4 Project |  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |  |        |   | OPERATING | ATM                 | AMB           | REF. PLAN DWG. | CJ—M15W14—PI—A30—030002—C |  | 25.03.31 | REVISED AS MARKED         |                                                                                                                     |  | 황래중 | 안남후 | 박동영   | JOB NO.    |  | 22330D                                                                                |  |                  |  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |  |        |   |           |                     |               | REF. P. & ID.  | CJ—M15W14—PR—A20—000115   |  | 25.01.24 | APPROVED FOR CONSTRUCTION |                                                                                                                     |  | 황래중 | 안남후 | 박동영   | LINE NO.   |  | 150A—D—ST1—N—9515                                                                     |  | SH'T NO.         |  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |  |        |   |           |                     |               |                |                           | REV. NO.                                                                              | DATE     | DESCRIPTION               |                                                                                                                     |  | BY  | CHK | APP'D |            |  | 01 / 01                                                                               |  | REV.<br>1        |  |

[illegible]

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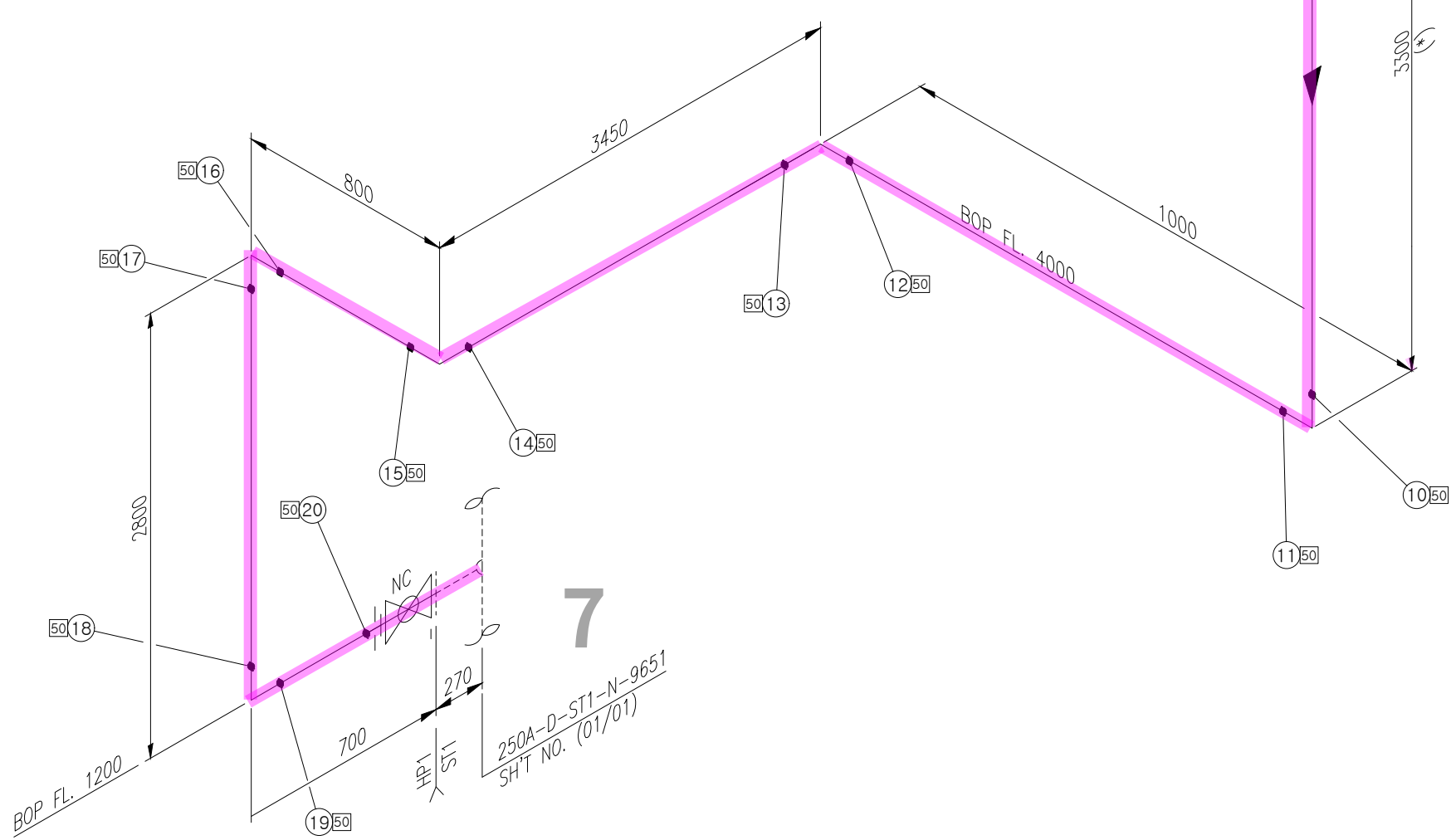
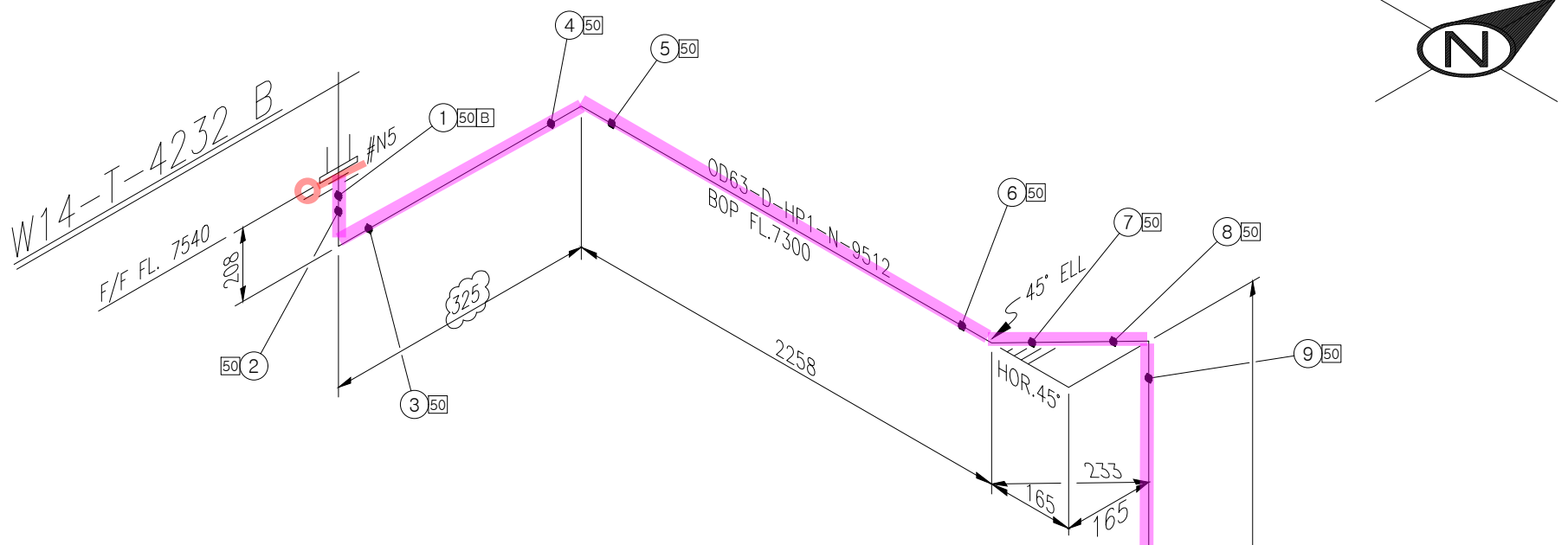
|                                                                                                  |                   |                  |      |
|--------------------------------------------------------------------------------------------------|-------------------|------------------|------|
| CONTRACTOR  |                   | PROJECT NAME     |      |
|                                                                                                  |                   | M15X Ph4 Project |      |
| COMMODITY D                                                                                      |                   | JOB NO.          |      |
|                                                                                                  |                   | 22330D           |      |
| LINE NO.                                                                                         | 250A-D-ST1-N-9651 | SH'T NO.         | REV. |
|                                                                                                  |                   | 01 / 01          | 1    |

[illegible]

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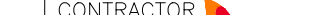
CLIENT  8

경고 : 본 도면은 SK HYNIX의 기밀비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유도자의 불법 사용으로 피해가 발생할 경우 유도자에게 민/형사상 책임을 요구할 수 있음



<NOTE>  
1. (\*) FIELD ADJUST

\* PIPE : P/P1 \* 90° ELBOW : E90 \* 45° ELBOW : E45 \* TEE : T \* RED.TEE : TR \* STUB-IN : STU \* REINFORCING PAD : RP \* CON.RED : RC \* ECC.RED : RE \* CON.SWAGE : RCS/RCS1 \* ECC.SWAGE : RES/RES1 \* NIPPLE : NP/NP1/NP2 \* CAP : CAP/CAP1 \* PLUG : PLG \* FULL COUPLING : FC \* HALF COUPLING : HF/HF1  
\* STUB-IN : STU \* SOCKET(HDPE/PVC) : SC \* SADDLE(HDPE) : SD \* FLANGE : F/F1 \* GASKET : G/G1/G2 \* BOLT & NUT : B/B1 \* BLIND FLANGE : FB/FB1 \* GATE VA. : VG/VG1/VG2 \* BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO \* BUTTERFLY VA. : VF/VF4 \* CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7  
\* GLOBE VA. : VL/VL1/VL2 \* DIAPHRAGM VA. : VD/VDLO \* DAMPER : VP \* QUICK COUPLING : SQC \* FLEXIBLE JOINT(STEEL) : FLXB \* FLEXIBLE JOINT(RUBBER) : FLXR \* FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL \* FLEXIBLE HOSE : FLXH \* STEAM TRAP : ST \* SPRAY NOZZLE : SPN \* EMERGENCY SHOWER : SSH  
\* STRAINER(Y-TYPE) : YSTR \* STRAINER(T-TYPE) : TSTR \* STRAINER(BUCKET-TYPE) : BSTR \* ANCHOR PLATE : STOP \* BIRD SCREEN : BS \* ISOLATION KIT : IK \* UNLOADING BOX : UB \* AIR TRAP : AT \* LOTO : LOTO \* SIGHT GLASS : SG \* PRESSURE GAUGE : PG \* SHILED : SH \* SPECIAL BOLT & NUT : BL/BL1

|            |        |   |           |                     |               |                                             |     |                                  |          |      |                           |     |     |                                                                                                                        |                   |                     |           |
|------------|--------|---|-----------|---------------------|---------------|---------------------------------------------|-----|----------------------------------|----------|------|---------------------------|-----|-----|------------------------------------------------------------------------------------------------------------------------|-------------------|---------------------|-----------|
| INSULATION | CODE   | N | CONDITION | PRESS.<br>(kg/cm2g) | TEMP.<br>(°C) | AREA                                        | W14 |                                  |          |      |                           |     |     | <div>CONTRACTOR</div> <div></div> | PROJECT NAME      |                     |           |
|            | THICK. | — | DESIGN    | 1.0                 | 60            | UNIT                                        | 3FA |                                  |          |      |                           |     |     |                                                                                                                        | M15X Ph4 Project  |                     |           |
|            |        |   | OPERATING | ATM                 | AMB           | REF. PLAN DWG.<br>CJ-M15W14-PI-A30-030002-E |     | <div>△</div> <div>REV. NO.</div> | 25.03.31 | DATE | APPROVED FOR CONSTRUCTION | 황래중 | 안남후 | 박동영                                                                                                                    | COMMODITY         | JOB NO.             |           |
|            |        |   |           |                     |               | REF. P & ID<br>CJ-M15W14-PR-A20-000115      |     |                                  |          |      |                           |     |     |                                                                                                                        | D                 | 22330D              |           |
|            |        |   |           |                     |               |                                             |     |                                  |          |      |                           |     |     | LINE NO.                                                                                                               | 0D63-D-HP1-N-9512 | SH'T NO.<br>01 / 01 | REV.<br>0 |